

The SMOG Model for Australia

Small Multivariate Output Gap Model — Specification and Implementation

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Abstract

This note documents the Australian SMOG model as implemented in `au_pipeline.py`. The model follows the Reserve Bank of Australia's small multivariate output gap framework, estimating potential output, the output gap and the NAIRU jointly as unobserved states. It is cast in state-space form and estimated by maximum likelihood using the Kalman filter.

1 Overview

The output gap is not observed. SMOG infers it by exploiting the fact that three observable series

- **Output** — actual GDP relative to its trend,
- **Unemployment** — via Okun's law $u_t - u_t^* = \beta \text{gap}_t + \varepsilon_t$, $\beta < 0$, u_t^* is the NAIRU
- **Inflation** — via the Phillips curve $\pi_t = \pi_t^e + \kappa \text{gap}_t + \varepsilon_t$, $\kappa > 0$.

The Kalman filter reconciles these three signals into a single coherent estimate of the gap, potential output and the NAIRU, weighting each signal by how informative it has proven historically.

2 Data and transformations

Implemented in `build_dataset()` function in code.

build_dataset() — full-sample transforms

```
1 def build_dataset(boost=1):
2     """Full-sample transforms exactly as in the EViews programme."""
3     raw = load_data()
4     d = pd.DataFrame(index=raw.index)
5
6     d["gdp"] = np.log(raw["real_gdp_non_farm_sa"])
7     cpi = raw["cpi_trimmed_spliced"]
8     d["inflation"] = pcy(cpi)
9     d["expectations"] = d["inflation"].rolling(4).mean() # @movav(.,4)
10    d["delta_nulc"] = pcy(raw["labour2"])
11    d["delta_4_pm"] = raw["import_price_deflator"] - raw["import_price_deflator"].shift(4)
12    d["unemployment"] = raw["unemployment"]
13    d["covid_d"] = raw["covid_d"]
14
15    # HP-filter initial values (EViews: hpf(lambda=1000, m=2, ic))
16    d["ham_u_star"] = hp_trend(d["unemployment"], boost=boost)
17    d["ham_y_star"] = hp_trend(d["gdp"], boost=boost)
18    d["ham_g_star"] = 100.0 * (d["ham_y_star"] / d["ham_y_star"].shift(1) - 1.0)
19    d["ham_gap"] = d["gdp"] - d["ham_y_star"]
20
21    # inflation-targeting dummy: 0 through 1993Q1, 1 after
22    d["d_it"] = (d.index > pd.Period("1993Q1", freq="Q")).astype(float)
23
24    d["y"] = d["gdp"]
```

```

25 d["pi"] = d["inflation"]
26 d["pi_e"] = d["expectations"]
27 d["u"] = d["unemployment"]
28 return d

```

Symbol	Code	Construction
y_t	y	log of real non-farm GDP (s.a.)
u_t	u	Unemployment rate
π_t	pi	Year-ended trimmed-mean CPI inflation
π_t^e	pi_e	4-quarter moving average of π_t
Δn_{ulc}_t	delta_nulc	Year-ended growth in nominal unit labour costs
Δ_{4pm}_t	delta_4_pm	4-quarter change in the import price deflator
D_t^{IT}	d_it	Inflation-targeting dummy: 0 to 1993Q1, 1 thereafter
covid _t	covid_d	COVID stringency index

Table 1: Observable series.

2.1 Initial values for the unobserved states

Starting values for the latent states come from a Hodrick–Prescott filter ($\lambda = 1000$, boosted once):

$$y_t^{*,HP} = HP(y_t), \quad \text{gap}_t^{HP} = y_t - y_t^{*,HP}, \quad (1)$$

$$u_t^{*,HP} = HP(u_t). \quad (2)$$

These are used only to initialise the filter and to generate starting values for the optimiser; they are not imposed on the final estimates.

3 Model specification

3.1 State vector

Seven states, of which three are economically meaningful and four are lags carried to make the system first-order Markov:

$$\xi_t = [\text{gap}_t \quad \text{gap}_{t-1} \quad \text{gap}_{t-2} \quad y_t^* \quad y_{t-1}^* \quad u_t^* \quad u_{t-1}^*]' \quad (3)$$

3.2 Measurement equations

Output. Actual output is potential output plus the gap, with a COVID adjustment:

$$y_t = y_t^* + \text{gap}_t + \phi_3 \text{covid}_t + \varepsilon_t^y \quad (4)$$

Okun’s law. Unemployment deviates from the NAIRU with the cycle, with persistence:

$$u_t = u_t^* + \lambda_3 \text{gap}_t + \rho (u_{t-1} - u_{t-1}^*) + \varepsilon_t^u \quad (5)$$

Phillips curve. Inflation responds to expectations, its own lags, import prices and the gap. The inflation-targeting dummy switches the specification between regimes:

$$\begin{aligned}
\pi_t = & (1 - \beta_1 - \beta_2) \pi_t^e D_t^{IT} \\
& + (1 - \beta_1 - \beta_2 - \beta_3 - \gamma) \pi_t^e (1 - D_t^{IT}) \\
& + \beta_1 \pi_{t-1} + \beta_2 \pi_{t-2} \\
& + (\beta_3 \pi_{t-3} + \gamma \Delta \text{nulc}_{t-1}) (1 - D_t^{IT}) \\
& + \lambda_1 \text{gap}_t + \psi \Delta_4 \text{pm}_{t-1} + \varepsilon_t^\pi
\end{aligned} \tag{6}$$

Before 1993 inflation carries more intrinsic persistence and unit labour costs matter; after the adoption of inflation targeting, expectations become anchored and those terms switch off.

3.3 State equations

The gap follows an AR(2); potential output is a random walk with drift; the NAIRU is a random walk:

$$\text{gap}_t = \phi_1 \text{gap}_{t-1} + \phi_2 \text{gap}_{t-2} + \varepsilon_t^{\text{gap}} \tag{7}$$

$$y_t^* = y_{t-1}^* + g^* + \varepsilon_t^{y^*} \tag{8}$$

$$u_t^* = u_{t-1}^* + \varepsilon_t^{u^*} \tag{9}$$

All disturbances are Gaussian, mutually uncorrelated, with variances $\epsilon_1^2, \dots, \epsilon_7^2$ estimated alongside the coefficients.

4 State-space representation

The model is written in the standard form

$$\text{(measurement)} \quad z_t = Z\xi_t + d_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, H) \tag{10}$$

$$\text{(transition)} \quad \xi_t = T\xi_{t-1} + c + R\eta_t, \quad \eta_t \sim N(0, Q) \tag{11}$$

where $z_t = [y_t, u_t, \pi_t]'$.

The split between Z and d_t is the key implementation detail:

- Z holds every term involving an *unobserved state* (`SmogModel.update()`),
- d_t holds every term that is *known data* — lags, dummies, expectations (`build_estimation_frames()`).

A single economic term may be split across both. In Okun's law, the persistence term $\rho(u_{t-1} - u_{t-1}^*)$ contributes ρu_{t-1} to d_t (observed) and $-\rho u_{t-1}^*$ to Z (a state).

Sample handling. Lags are computed *within* the estimation window rather than from the full history, so the first observations have genuinely missing regressors. These are treated as partially observed: the affected measurement equation is dropped from the likelihood for that period only.

5 Estimation

5.1 Starting values

Implemented in `ols_start_params()`. Each equation is estimated separately with the HP-filtered states substituted for the true latent states:

Equation	Method	Parameters
Output	OLS	ϕ_3
Okun	OLS	$\lambda_3, \rho, \epsilon_2$
Phillips	NLS	$\beta_1, \beta_2, \beta_3, \gamma, \lambda_1, \psi, \epsilon_3$
Gap AR(2)	OLS	$\phi_1, \phi_2, \epsilon_5$
Potential	Mean of $\Delta y^{*,HP}$	g^*, ϵ_6

Table 2: Starting-value regressions. The Phillips curve is non-linear in the coefficients because they are restricted to sum appropriately across regimes.

5.2 Maximum likelihood

The likelihood is evaluated by the Kalman filter and maximised numerically. The optimiser alternates BFGS with derivative-free methods, since the likelihood surface is flat in some directions.

The filter is initialised with the HP-based values as the mean of the state at $t = 1$, with a zero prior covariance.

6 Output

The Kalman smoother produces full-sample estimates of every state. The reported series are:

- **Output gap:** $100 \times \text{gap}_{t|T}$, per cent deviation of output from potential,
- **Potential output:** $y_{t|T}^*$,
- **NAIRU:** $u_{t|T}^*$.

Smoothed estimates use the entire sample and so revise as data accrues; filtered estimates use only information available at each date and are the appropriate real-time measure.

7 Implementation notes

- Estimation sample: 1980Q1–2023Q4.
- Variance parameters are estimated as standard deviations and squared internally, so their sign is arbitrary.
- The unemployment series is sourced from the maintained inputs workbook: the estimation workbook’s column is date-shifted by a broken data refresh.